

AI Campus Route Navigator — Assignment X_03

AI/ML Laboratory | B.Tech. 5th Semester

1. Objective

The project converts the supplied CU Technology Campus map into a weighted graph and compares PATHFINDER (Greedy Best-First Search) with ORBIT (A* Search). The comparison considers route cost, nodes explored, execution time, and the CSE-zone routing restriction.

2. Implementation

The graph is stored in campus.json. Search is implemented from scratch using Python's heapq priority queue. PATHFINDER uses $f(n)=h(n)$, while ORBIT uses $f(n)=g(n)+h(n)$. Both agents use the same straight-line map heuristic.

3. CSE Constraint

CSE Laboratory, CSE_AKC Seminar Hall, and CSE_Reflxon Room form the CSE zone. Entry must pass through a Tower 2 entry and Lift Area; while inside, only CSE locations may be visited until returning to Lift Area. The constraint is enforced during neighbour generation.

4. Experimental Results

Route	PF cost	A* cost	PF nodes	A* nodes	Different?
Reception → Library	465	425	5	11	Yes
Canteen → New Building 2	255	255	3	7	No
Gate 1 → CSE Laboratory	575	575	8	15	No
Auditorium → CSE_AKC	325	325	5	12	No
Library → Canteen	480	440	5	15	Yes
Gate 4 → New Building 2	350	350	4	5	No
Power Area → Library	390	350	4	11	Yes
Gate 2 → CSE_Reflxon	310	310	5	8	No

5. Observations

Across the eight tested pairs, PATHFINDER averaged 393.75 m and 4.88 explored nodes; ORBIT averaged 378.75 m and 10.50 explored nodes. The agents chose different routes for three pairs, and ORBIT produced the lower-cost route in all three. This demonstrates that a location that looks closest according to $h(n)$ does not always produce the lowest-cost complete route.

6. Conclusion

Greedy Best-First Search is lighter on the tested graph but can commit to a locally attractive route. A* performs more search because it includes accumulated cost $g(n)$, and in the differing cases it found the lower-cost route. The CSE routing rule also shows that domain constraints must be incorporated into state transitions rather than checked only after a route is produced.

All numerical results in this report were generated from the submitted implementation and saved to results/results.csv.